Lecture 7: Causality

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Applied Analytical Statistics

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Addendum | Testing for overdispersion in Poisson regression

Poisson regression (useful for count outcomes) assumes that variance = mean.

$\log(\lambda_i) = \mathbf{X}_i\beta$ where $Y_i \sim \text{Poisson}(\lambda_i)$ with $\lambda_i = E(Y_i|X) = \text{Var}(Y_i|X)$

When variance > mean, there is **overdispersion**, which leads to incorrect SEs.

We test for overdispersion by comparing log-likelihoods ℓ of our model vs "perfect" model:

$$\hat{\phi} = \frac{\text{Residual Deviance}}{\text{Degrees of Freedom}}$$

where

$$\text{Residual Deviance} = 2(\ell_{\text{saturated}} - \ell_{\text{model}})$$

has one parameter per observation, fits data perfectly

When $\hat{\phi} > 1$ this is evidence of overdispersion. Rule of thumb: $\hat{\phi} > 1.5$ is concerning.

To fix: use robust ("sandwich") SEs, or use other model like negative binomial

Addendum | Average Marginal Effect in logistic regression

Logistic model

$$\log\left(\frac{p_i}{1-p_i}\right) = X_i\boldsymbol{\beta} \quad \equiv \quad p_i = \frac{e^{X_i\boldsymbol{\beta}}}{1+e^{X_i\boldsymbol{\beta}}}$$

Marginal effect (ME)

$$\frac{\partial p}{\partial X_{ik}} = \beta_k p_i(1-p_i)$$

on the probability scale

Average marginal effect (AME)

$$\mathrm{AME}_k = \frac{1}{n}\sum_{i=1}^{n} \beta_k \; p_i(1-p_i)$$

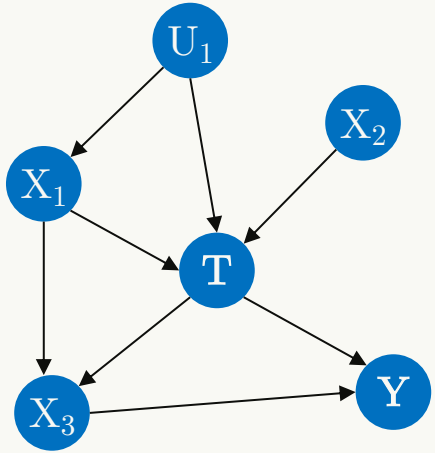
ME averaged over all observations

The average change in predicted probability associated with a one-unit increase in X_k, holding other variables constant.

Example: $p_i = \mathrm{logit}^{-1}(2 + 0.1X_1 - 1.3X_2)$

1. Predict p_i for every observation i.
2. Compute ME for β_2 for each i.
3. Average MEs for β_2 over all observations.

Plan for today | Causality



Today, we learn what it takes to move from describing associations to **identifying causal effects**.

We cover **potential outcomes**: reasoning about counterfactuals to make causal claims.

We cover causal diagrams: how to read them, work with them, draw your own.

We will finish with a class activity on causal diagrams to help you with your summative.

Working example | AI use and productivity

We now have the tools to answer **descriptive** RQs like:

hypothesis tests

multivariate regression

RQ: Is using AI associated with being more productive?

But the more interesting RQs are **causal**:

RQ: Does using AI make workers more productive?

Causal evidence produces **actionable insights** for decisions like:

- Should AI adoption be encouraged?
- Who benefits from AI adoption? Who may be harmed?

Descriptive evidence can be misleading!

e.g. OVB: worker skill explains AI use and productivity

Potential outcomes | Notation

Let $Y_i \in \mathbb{R}^+$ be productivity of worker i and $T_i \in \{0,1\}$ indicate whether worker i uses AI:

$$Y_i = \begin{cases} Y_i^0 \text{ if } T_i = 0 & \rightarrow \text{worker does not use AI} \\ Y_i^1 \text{ if } T_i = 1 & \rightarrow \underline{\text{same}} \text{ worker uses AI} \end{cases}$$

$$Y_i = Y_i^0 + T_i(Y_i^1 - Y_i^0)$$

potential outcomes

Now we can define **causal effects** as the difference between two states of the world:

For the individual: $\delta_i = Y_i^1 - Y_i^0$, the unit-level treatment effect.

Averaged across workers: $\text{ATE} = E[\delta_i] = E[Y_i^1] - E[Y_i^0]$, the average treatment effect.

The **Fundamental Problem of Causal Inference** is that we can never observe both states of the world at the same time.

or

Potential outcomes | Connection to regression

Our goal is to recover the causal effect $\text{ATE} = E[Y_i^1] - E[Y_i^0]$ from observed data.

Knowing outcome Y_i and treatment T_i for each i, we can write a simple linear regression:

$E[Y_i|T_i] = \beta_0 + \beta_1 T_i$ which implies $\beta_1 = E[Y_i|T_i = 1] - E[Y_i|T_i = 0]$

Estimation is the problem of learning the population coefficient β_1 from data.
This is **statistical inference**: under which conditions does $\hat{\beta}_1 = \beta_1$?

Identification is the problem of interpreting coefficients as causal effects.
This is **causal inference**: under which conditions does $\beta_1 = \text{ATE}$?

$$\hat{\beta}_1 \xrightarrow{\text{estimation}} \beta_1 \xrightarrow{\text{identification}} \text{ATE}$$

our focus today

Potential outcomes | Identification of ATE via regression

When does $\beta_1 = E[Y_i|T_i = 1] - E[Y_i|T_i = 0]$ equal $ATE = E[Y_i^1] - E[Y_i^0]$?

If and only if $E[Y_i|T_i = 1] = E[Y_i^1]$ and $E[Y_i|T_i = 0] = E[Y_i^0]$

1) This requires consistency: observed outcomes equal relevant potential outcomes:

$Y_i = Y_i^0 + T_i(Y_i^1 - Y_i^0)$ so that $E[Y_i|T_i = 1] = E[Y_i^1|T_i = 1]$

Violated when there are multiple versions of treatment or spillover effects across units.

2) This requires exchangeability: assigned treatment T_i is independent of potential outcomes:

$T_i \perp (Y_i^0, Y_i^1)$ so that $E[Y_i^1|T_i = 1] = E[Y_i^1|T_i = 0] = E[Y_i^1]$

Violated for confounding and selection into treatment.

Potential outcomes | Selection bias decomposition

To better understand exchangeability, we can expand the coefficient equation:

$$\begin{aligned}\beta_1 &= E[Y_i|T_i = 1] - E[Y_i|T_i = 0] \\ &= \underbrace{(E[Y_i^1|T_i = 1] - E[Y_i^0|T_i = 1])}_{\text{ATT}} + \underbrace{(E[Y_i^0|T_i = 1] - E[Y_i^0|T_i = 0])}_{\text{selection bias}}\end{aligned}$$

under assumption of consistency

ATT

What is the average treatment effect among treated units?

selection bias

How different would treated units have been even without treatment?

When exchangeability holds, $\beta_1 = \text{ATE}$ because:

$T_i \perp (Y_i^0, Y_i^1)$

implies $\underline{E[Y_i^0|T_i = 1] = E[Y_i^0|T_i = 0]}$ $\rightarrow$ no selection bias

implies $\underline{E[Y_i^1|T_i = 1] - E[Y_i^0|T_i = 1]} = E[Y_i^1] - E[Y_i^0]$ $\rightarrow$ ATT = ATE

Potential outcomes | Violations of exchangeability

$$\beta_1 = \underbrace{(E[Y_i^1|T_i = 1] - E[Y_i^0|T_i = 1])}_{\text{ATT}} + \underbrace{(E[Y_i^0|T_i = 1] - E[Y_i^0|T_i = 0])}_{\text{selection bias}}$$

RQ: Does using AI make workers more productive?

Suppose skilled workers are more productive and more likely to adopt AI.
This violates exchangeability: **potential outcomes systematically differ between groups**.

$E[Y_i^0|T_i = 1] > E[Y_i^0|T_i = 0]$ even without AI, thus: $T_i \not\perp (Y_i^0, Y_i^1)$

This creates selection bias, so that $\beta_1 \neq \text{ATT}$.

🤔 What can we do about this?

Potential outcomes | The power of randomisation

We can **guarantee exchangeability by design** if we **randomly assign treatments**.

$T_i \perp (Y_i^0, Y_i^1)$ always holds if we flip a coin to determine T_i

Randomised
Controlled
Trial

Treatment is statistically unrelated to all pre-treatment characteristics.
Any difference we observe after treatment is attributable to treatment in expectation.

RQ: Does using AI make workers more productive?

If we randomly assign workers to use AI or not, we can rule out that worker skill (or any other unobserved variable) systematically influences whether workers use AI or not.

🤔 But what if all we have is observational data?

Potential outcomes | Conditional exchangeability

In observational data, treatment is rarely independent of potential outcomes.

However, we may have **conditional exchangeability** $T_i \perp (Y_i^0, Y_i^1) \mid X_i$

→ Within subgroups defined by control variables X, assignment of T is as good as random.

$E[Y_i^0 | T_i = 1, X_i] = E[Y_i^0 | T_i = 0, X_i]$ and $E[Y_i^1 | T_i = 1, X_i] = E[Y_i^1 | T_i = 0, X_i]$

IF conditional exchangeability holds, and we fit a multivariate regression:

$E[Y_i | T_i, X_i] = \beta_0 + \beta_1 T_i + \beta_2 X_i$ then $\beta_1 = \text{ATE}$

→ Multivariate regression can, in principle, recover a causal effect from observational data.

🤔 But how do we decide what to control for?

Identification | Data generating processes (DGPs)

As scientists, we (mostly) believe there are **underlying causal mechanisms** that govern how the world works, and we can learn about these mechanisms from empirical data.

Every dataset is the output of a **data generating process** (DGP) that describes:

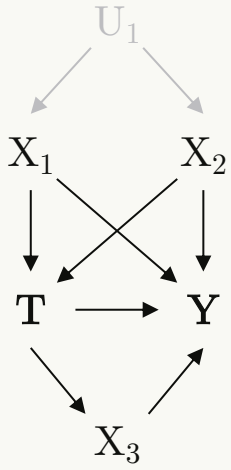
theory about reality

- Which variables cause which other variables
- The temporal order of events
- Which variables are observed vs unobserved
- The functional relationships between variables

In order to **justify claims about causal effects based on observational data**, we need to **make assumptions about the DGP** that generated the data we observe.

β_1 = ATE requires (conditional) exchangeability & (conditional) exchangeability depends on the DGP!

Causal diagrams | Directed Acyclic Graphs



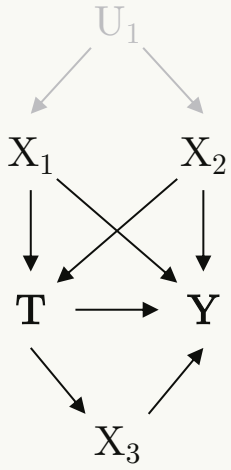
A causal diagram is a graphical representation of an assumed DGP.

Directed Acyclic Graphs (DAGs) help us reason about identification.
Included variables = relevant to T or Y, directly or indirectly
Included arrows = direct causal effects

DAGs can include unobserved variables.

Omissions from the DAG are also assumptions:
Omitted variables = not relevant to T or Y
Omitted arrows = no direct causal effects

Causal diagrams | Terminology



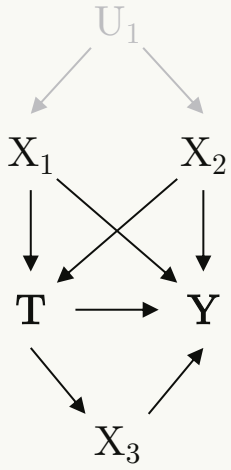
Variables are **nodes**.

Nodes can be **descendants** or **ancestors** of other nodes:
X_1 is a descendant of U_1. X_3 is a descendant of X_1, etc.
U_1 is an ancestor of X_1. X_2 is an ancestor of X_3, etc.

Parents are direct causes of all their **children**.
X_1 is a direct cause of T and Y. U_1 is a direct cause of X_1 and X_2, etc.

A **path** is any sequence of connected edges, ignoring arrow direction.
$U_1 \to X_1 \to T \to Y$ is a path.
$X_1 \leftarrow U_1 \to X_2$ is also a path.

Causal diagrams | Front doors and back doors



RQ: What is the effect of treatment **T** on outcome **Y**?

To answer the RQ, we first list all paths that connect T and Y:

$T \rightarrow Y$

$T \rightarrow X_3 \rightarrow Y$

front door paths begin with an arrow away from T

"good paths" if relevant to RQ

$T \leftarrow X_1 \rightarrow Y$

$T \leftarrow X_2 \rightarrow Y$

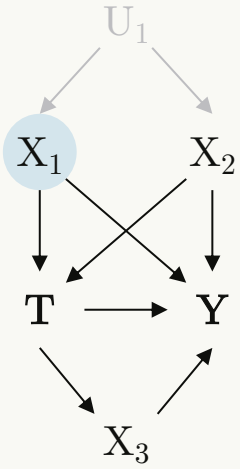
$T \leftarrow X_1 \leftarrow U_1 \rightarrow X_2 \rightarrow Y$

$T \leftarrow X_2 \leftarrow U_1 \rightarrow X_1 \rightarrow Y$

back door paths begin with an arrow towards T

"bad paths" because they offer alternative explanations for association in T and Y

Causal diagrams | Open and closed paths



A causal path is **closed** if at least one of the variables along the path has no variation / is not free to vary.

By **conditioning on a variable**, we can close all paths through it.*

conditioning on X_1 = comparing units at the same value of X_1

$T \rightarrow Y$

$T \rightarrow X_3 \rightarrow Y$

$T \leftarrow X_1 \rightarrow Y$

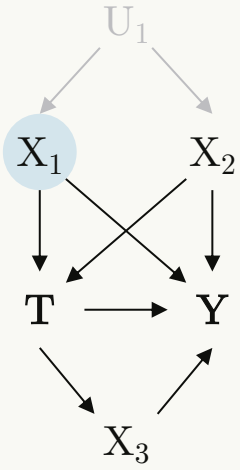
$T \leftarrow X_2 \rightarrow Y$

$T \leftarrow X_1 \leftarrow U_1 \rightarrow X_2 \rightarrow Y$

$T \leftarrow X_2 \leftarrow U_1 \rightarrow X_1 \rightarrow Y$

*with one important exception, will cover shortly

Causal diagrams | How to condition

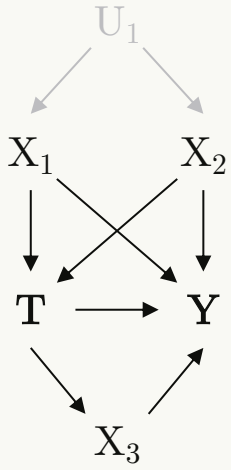


Conditioning on X_1 means comparing units at the same value of X_1.

We can condition by **partitioning the data** on X_1.
Within each subgroup, there is no variation in X_1.

We can condition by **controlling for X_1** in a regression.
We use only variation in T that is independent of $\mathbf{X_1}$.

Causal diagrams | The backdoor criterion



To figure out what to control for, we can use the **backdoor criterion**.

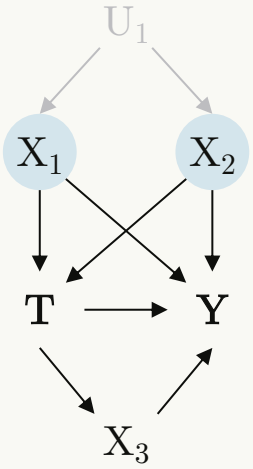
A set of variables Z satisfies the **backdoor criterion** relative to (T,Y) if:

1. No element of Z is a descendant of T.
2. Conditioning on Z blocks every backdoor path between T and Y.

If **Z** satisfies the backdoor criterion, then $T \perp (Y^0, Y^1) \mid Z$

conditional exchangeability!

Causal diagrams | The backdoor criterion in action

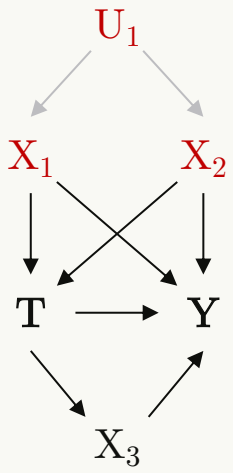


In our case, only $\mathbf{Z} = \{X_1, X_2\}$ satisfies the backdoor criterion.

$T \to Y$
$T \to X_3 \to Y$
$T \leftarrow X_1 \to Y$
$T \leftarrow X_2 \to Y$
$T \leftarrow X_1 \leftarrow U_1 \to X_2 \to Y$
$T \leftarrow X_2 \leftarrow U_1 \to X_1 \to Y$

when conditioning on Z,
only "good paths" remain open!

Causal diagrams | Confounders



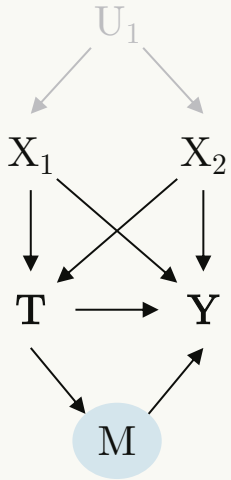
A variable is a **confounder** for the effect of T on Y if it is:

- A direct or indirect cause of T, and
- A direct or indirect cause of Y, and
- Not a descendant of T.

The **backdoor criterion** tells us which variables to condition on to eliminate the influence of confounders.

example: UsesAI $\leftarrow$ Skill $\rightarrow$ Productivity

Causal diagrams | Mediators



A **mediator M** is a variable that lies on a directed path from T to Y.

The **total effect** of T on Y can be decomposed as:

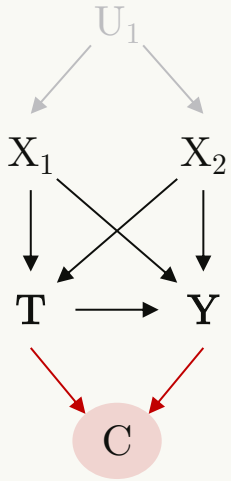
direct effect: $T \to Y$

+ mediated effect: $T \to M \to Y$

By controlling for M, we rule out causal effects transmitted through M. This may or may not be desirable, depending on our RQ.

example: UsesAI → TimePerSubTask → Productivity

Causal diagrams | Collider bias



A collider C is a variable that is caused by both T and Y.

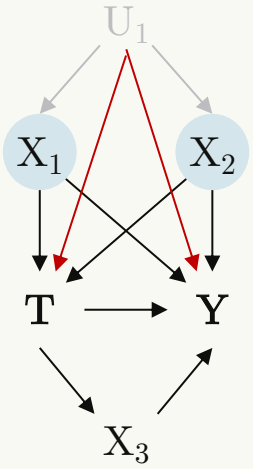
Paths through colliders are closed by default: $T \rightarrow C \leftarrow Y$

Conditioning on a collider opens the path.

Conditioning colliders creates statistical dependence between T and Y.
If we condition on C, learning T gives information about Y.

example: UsesAI $\rightarrow$ Promotion $\leftarrow$ Productivity

Causal diagrams | Unobserved confounding



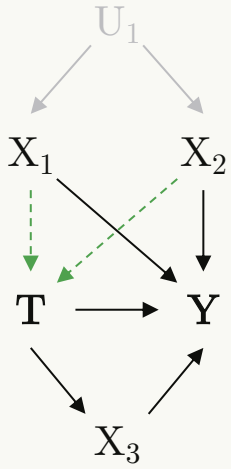
There may be backdoor paths through unobserved variables.

$T \leftarrow U_1 \rightarrow Y$
$T \leftarrow X_1 \leftarrow U_1 \rightarrow Y$
$T \leftarrow X_2 \leftarrow U_1 \rightarrow Y$

If so, we **cannot close these backdoor paths through conditioning**.

example: UsesAI $\rightarrow$ Promotion $\leftarrow$ Productivity

Causal diagrams | Randomisation



Random assignment of T removes <u>direct causal effects</u> on T.
i.e. incoming arrows

$T \to Y$
$T \to X_3 \to Y$ only front door paths remain!

Randomisation guarantees exchangeability $T \perp (Y^0, Y^1)$
so that conditioning on controls is no longer necessary.

Drawing DAGs | Selecting variables for inclusion

Drawing a causal diagram means formalising our idea of the data generating process. DGPs in the real world are complex. We focus only on variables relevant to the RQ.

RQ: Does **using AI** make workers more **productive**?

What may cause **T = UsesAI**?

Age
Skill
Education
AccessToAI
TechAttitude
Role

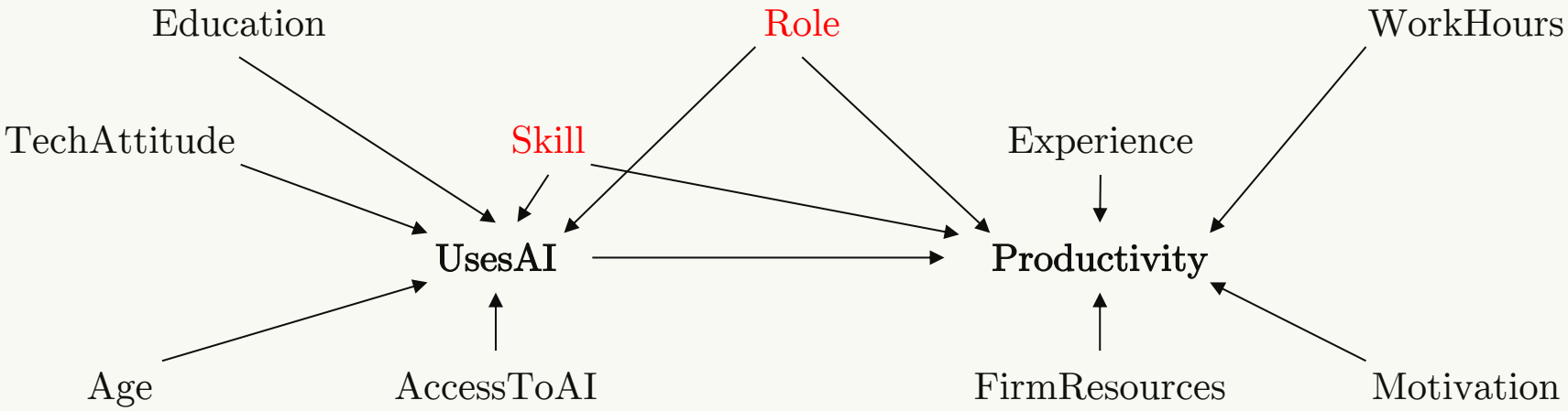
What may cause **Y = Productivity?**

Skill
Motivation
Experience
Role
WorkHours
FirmResources

What may cause both **T and Y?**

Drawing DAGs | Direct effects on T and Y

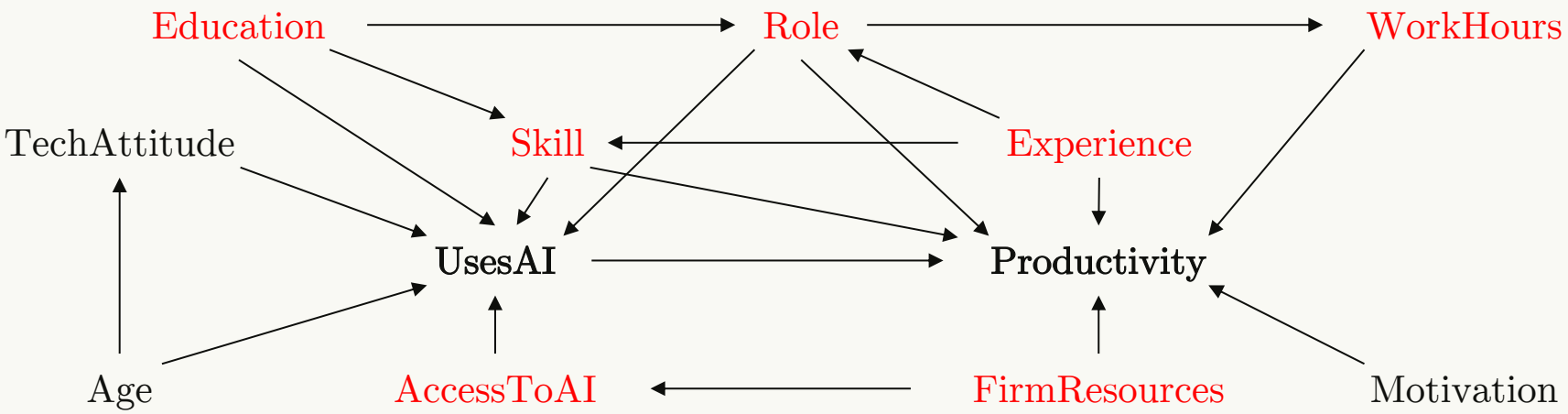
First, we **add the direct effects on T and Y** that we used to motivate variable selection.



Already, we can see two **confounders** on backdoor paths: variables that cause T and Y.

Drawing DAGs | Effects between variables

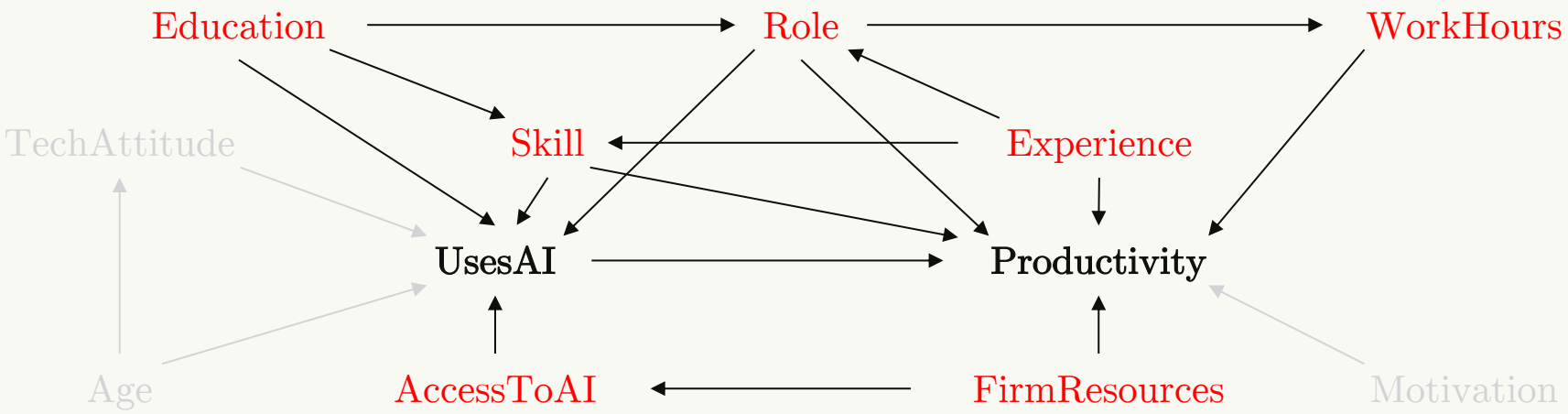
Second, we **add the effects between all other variables** that we include in our DAG.



Now, we see that most variables in our DAG are **confounders** (directly or indirectly).

Drawing DAGs | Simplification by irrelevance

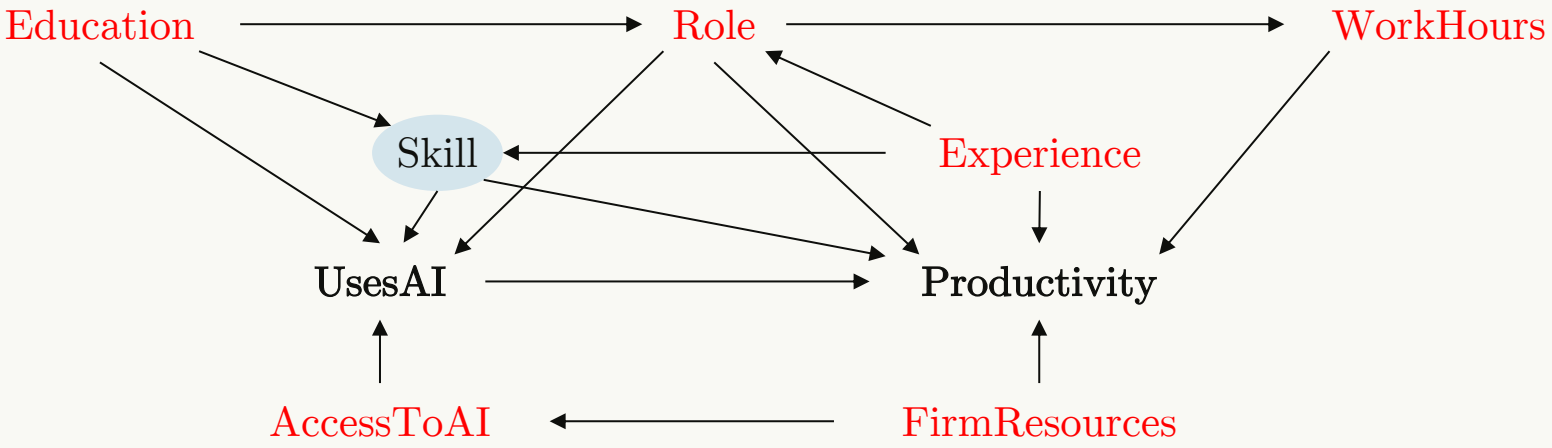
Next, we can remove **irrelevant** variables, i.e. those not on any path between T and Y.



In principle, we could also collapse redundant variables that have the same in/out arrows.

Drawing DAGs | Closing causal paths by conditioning

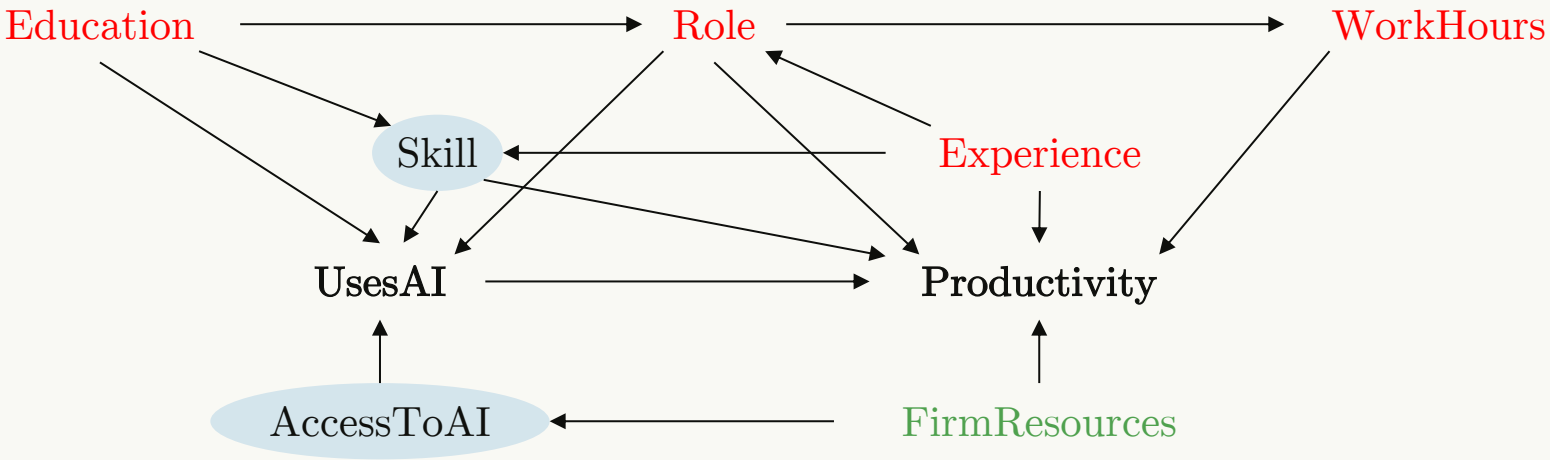
Finally, we can reason through the consequences of **conditioning on (sets of) variables**.



Our goal is to eliminate the influence of all **confounders**. Effectiveness of conditioning varies.

Drawing DAGs | Closing causal paths by conditioning

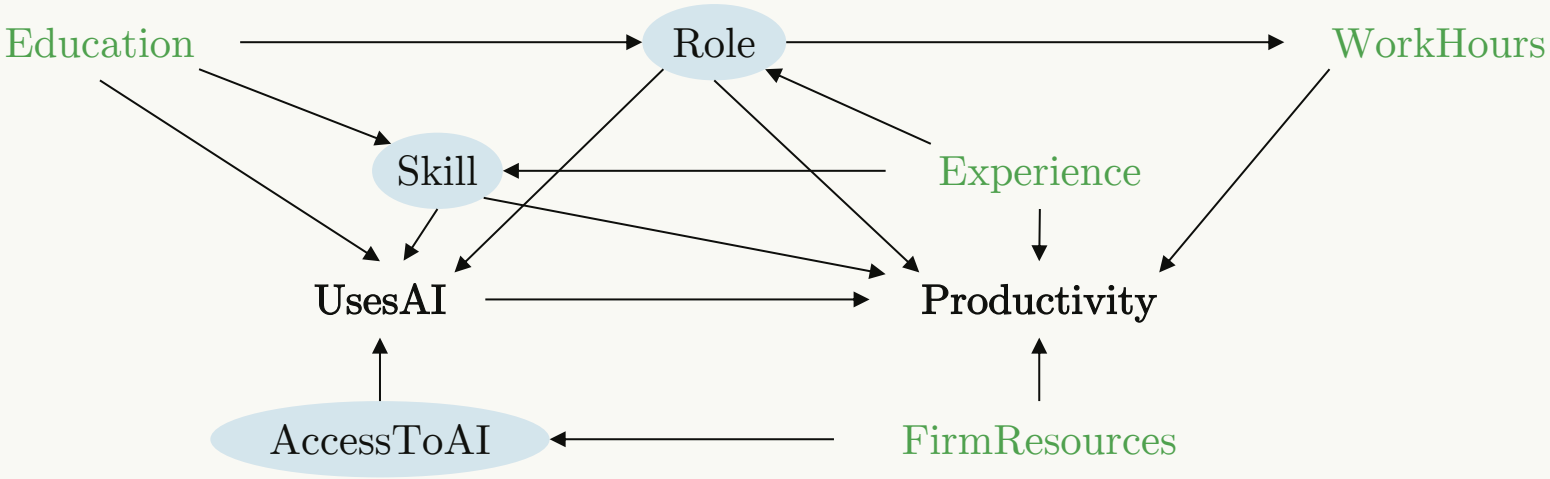
Finally, we can reason through the consequences of **conditioning on (sets of) variables**.



Our goal is to eliminate the influence of all **confounders**. Effectiveness of conditioning varies.

Drawing DAGs | Applying the backdoor criterion

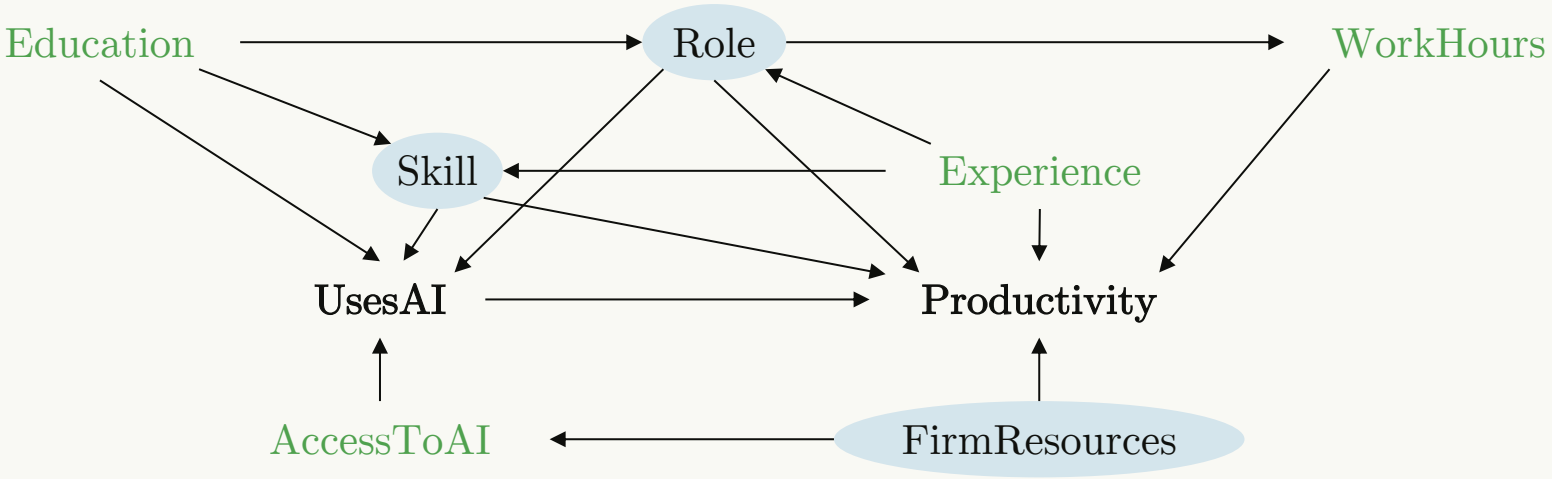
The set of variables $\mathbf{Z} = \{\text{Role, Skill, AccessToAI}\}$ satisfies the backdoor criterion.



Assuming our DAG is true, we obtain conditional exchangeability: $T \perp (Y^0, Y^1) \mid Z$

Drawing DAGs | Applying the backdoor criterion

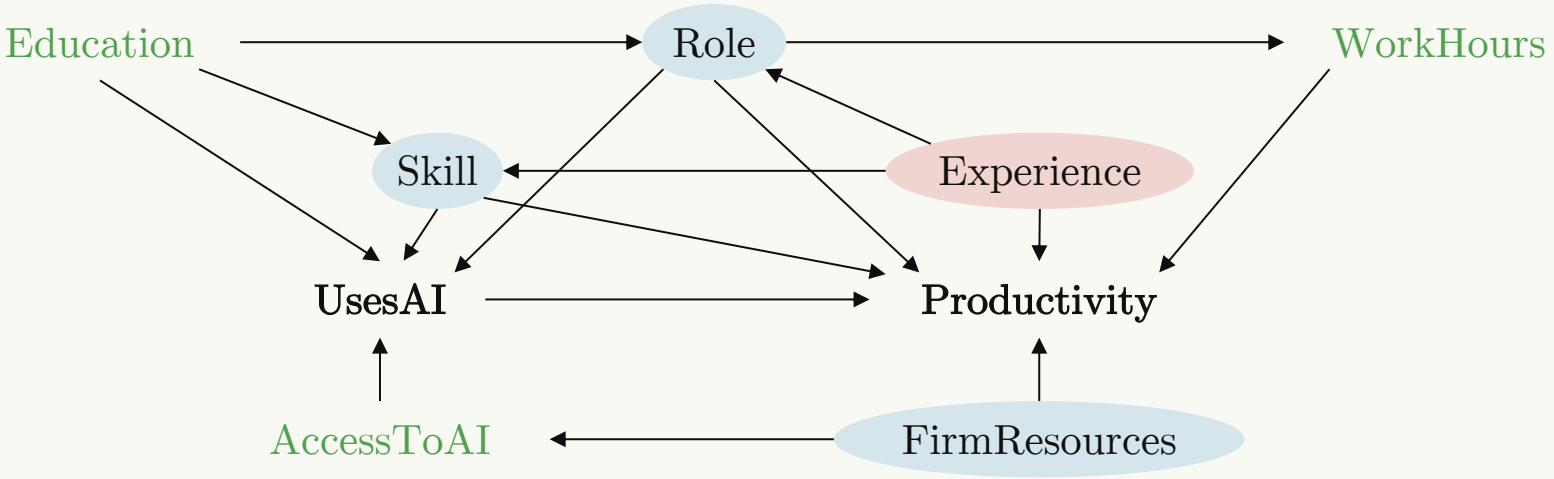
More than one set of variables **Z** may satisfy the backdoor criterion.



Common direct causes of T and Y (here: Skill and Role) must always be included in Z.

Drawing DAGs | Beware of overcontrolling

After conditioning on Skill and Role, we do not need to condition on Experience anymore.



Overcontrolling <u>does not threaten identification</u> but <u>may harm estimation</u>.
(if the extra control variable is neither a collider nor descendant of T)

(FWL: control removes usable variation in T)

Drawing DAGs | My personal take

DAGs are useful as a **tool for helping you reason about identification in the real world**. They formalise assumptions and allow us to assess whether a causal effect is identifiable.

All DAGs are simplifications. Drawing the DAG for your RQ forces you to think about which causal paths are relevant to your RQ and important / strong enough to be included.

Perfect conditional exchangeability is a myth. You can identify all relevant controls under the assumption of a DAG, but even the best DAGs are simplified models of the real world.

Strength of identification lies on a spectrum. Only specific experimental designs allow for ”clean” causal claims. Conditioning makes causal claims more or less plausible.

Class activity | Draw your own DAG

You are not expected to make causal claims in your summative
BUT you should still think about the DGP that produced your data.

When designing your analysis, think of confounders and controls.
When discussing your results, highlight omitted variables as limitations.

1. Identify treatment T and outcome Y in your RQ.
2. Find 3 variables that cause T and 3 variables that cause Y.
3. Try to find at least one variable that causes both T and Y.
4. Complete your DAG by drawing all relevant causal paths.

Please work in the shared Canva whiteboard (Canvas → Week 7 Class Activity)

Recap | Key takeaways from Week 7

Causal claims require identifying assumptions.
Before estimating a causal effect, we must justify it is identifiable from the observed data.

Potential outcomes formalise causal effects.
Causal effects are concerned with counterfactuals. Identification requires exchangeability.

Directed Acyclic Graphs (DAGs) help us reason about identification.
DAGs specify assumptions about the data-generating process, including confounders.

The DAG backdoor criterion tells us which variables to condition on.
We can establish conditional exchangeability by blocking all backdoor paths.